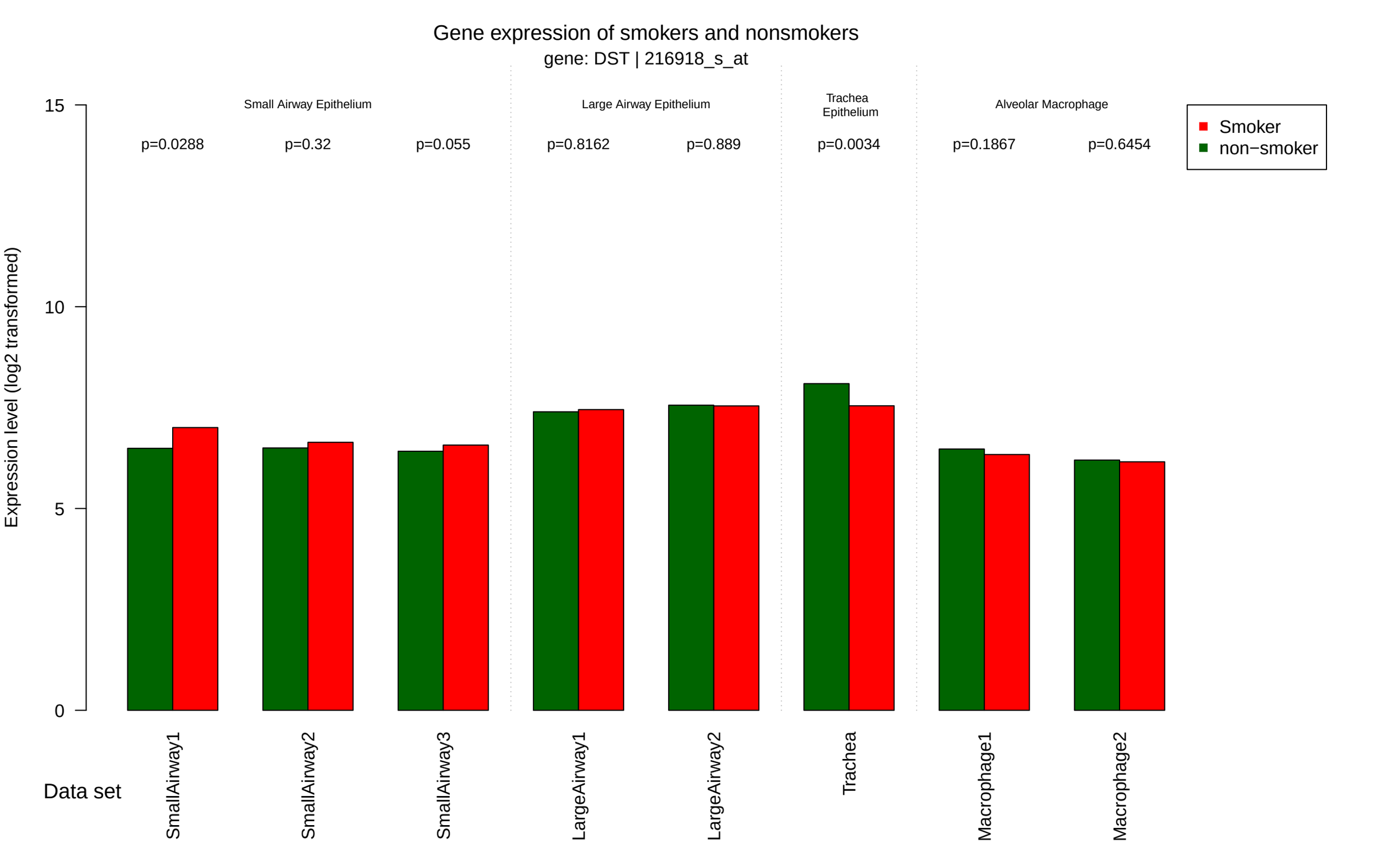
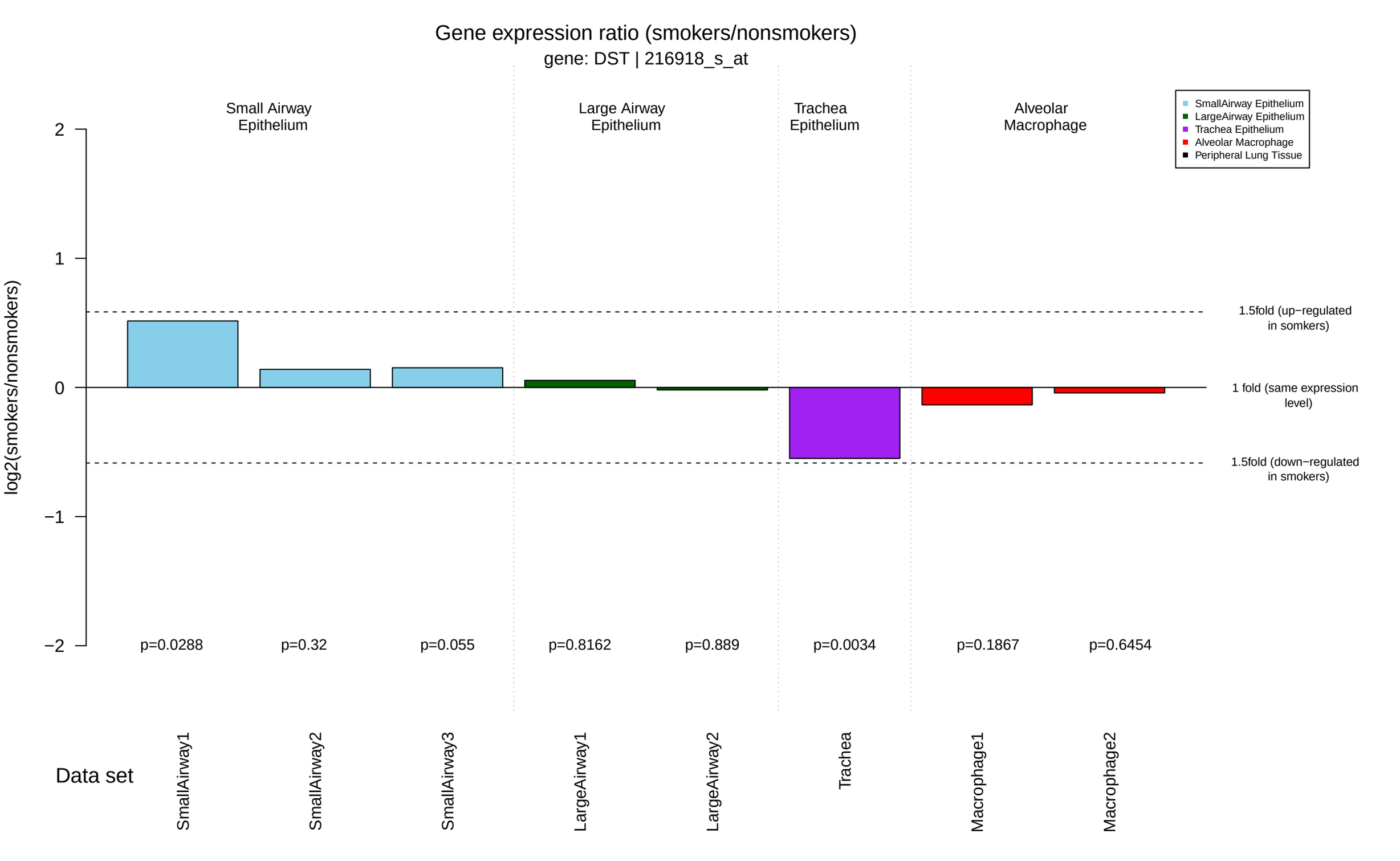
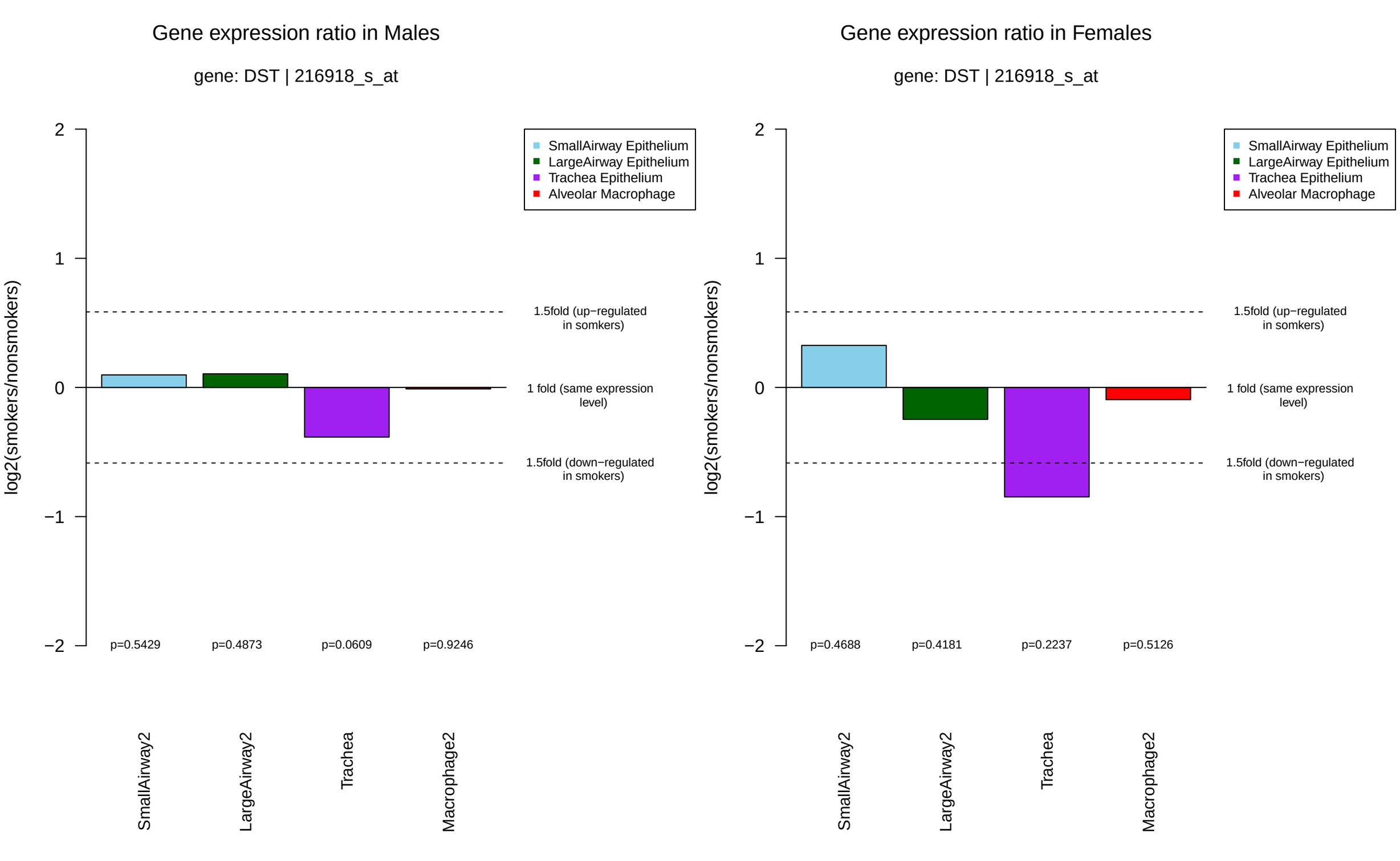
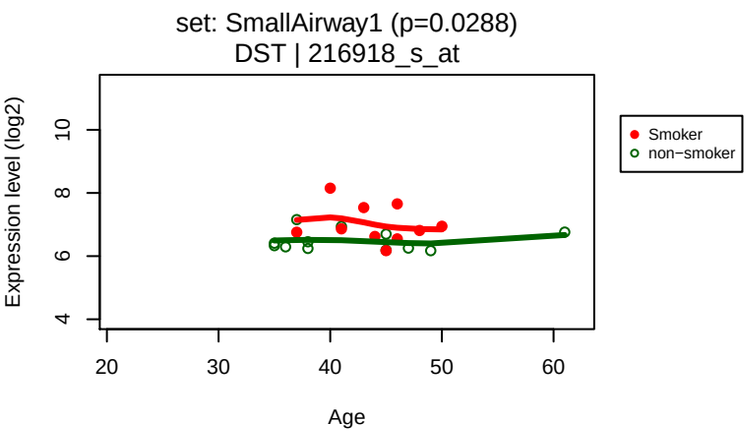
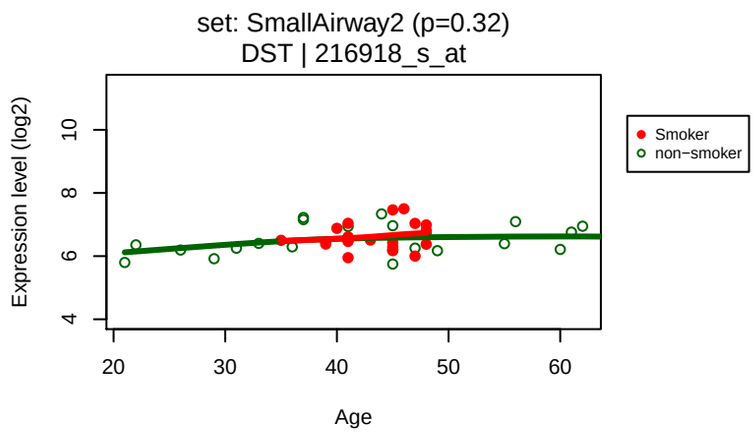
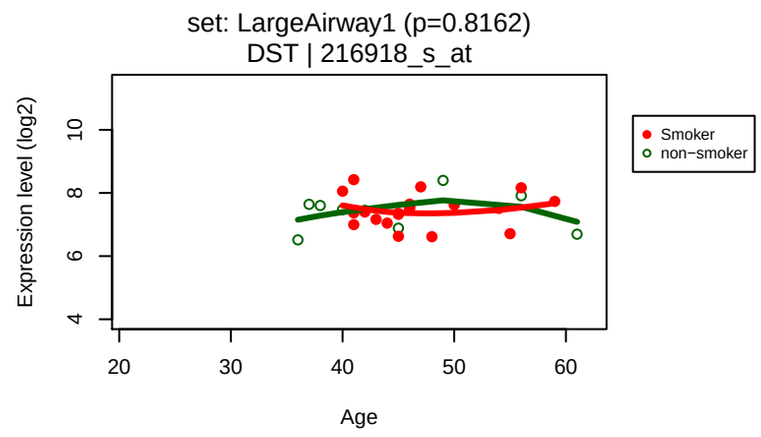

Gene expression of smokers and nonsmokers
gene: DST | 216918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0288
p=0.32
p=0.055
p=0.8162
p=0.889
p=0.0034
p=0.1867
p=0.6454
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

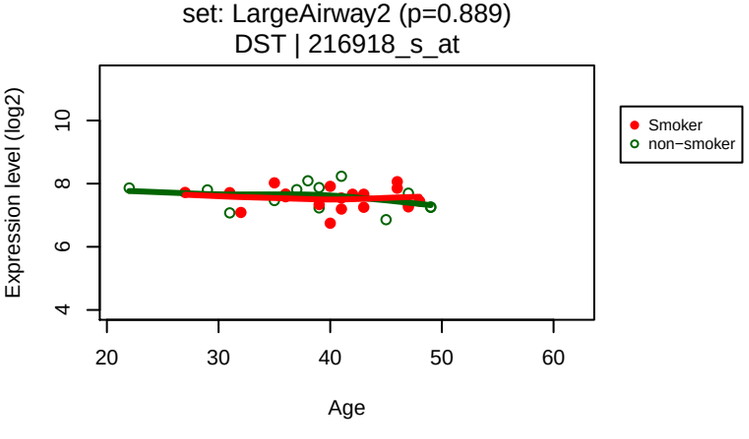

Gene expression ratio (smokers/nonsmokers)
gene: DST | 216918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0288
p=0.32
p=0.055
p=0.8162
p=0.889
p=0.0034
p=0.1867
p=0.6454
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

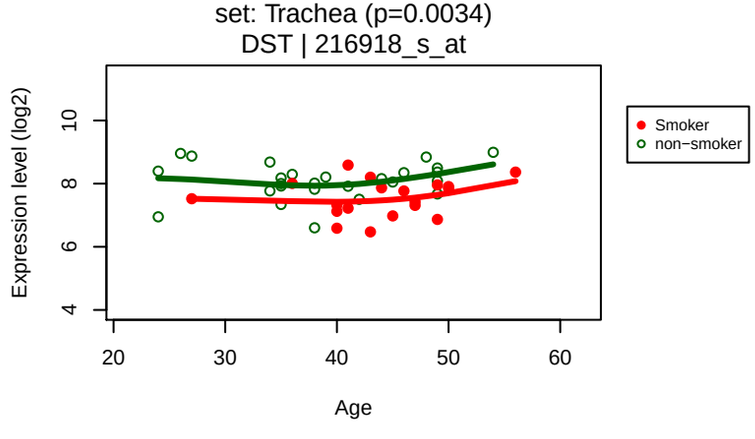

Gene expression ratio in Males
gene: DST | 216918_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5429
p=0.4873
p=0.0609
p=0.9246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DST | 216918_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4688
p=0.4181
p=0.2237
p=0.5126
SmallAirway2
LargeAirway2
Trachea
Macrophage2

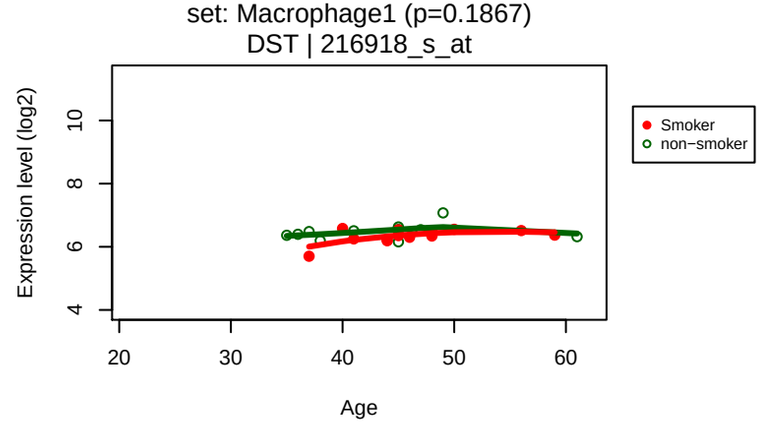

set: SmallAirway1 (p=0.0288)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker

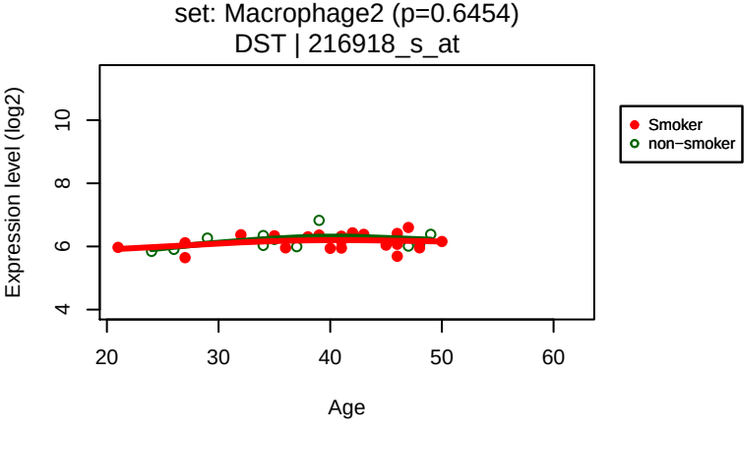

set: SmallAirway2 (p=0.32)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker

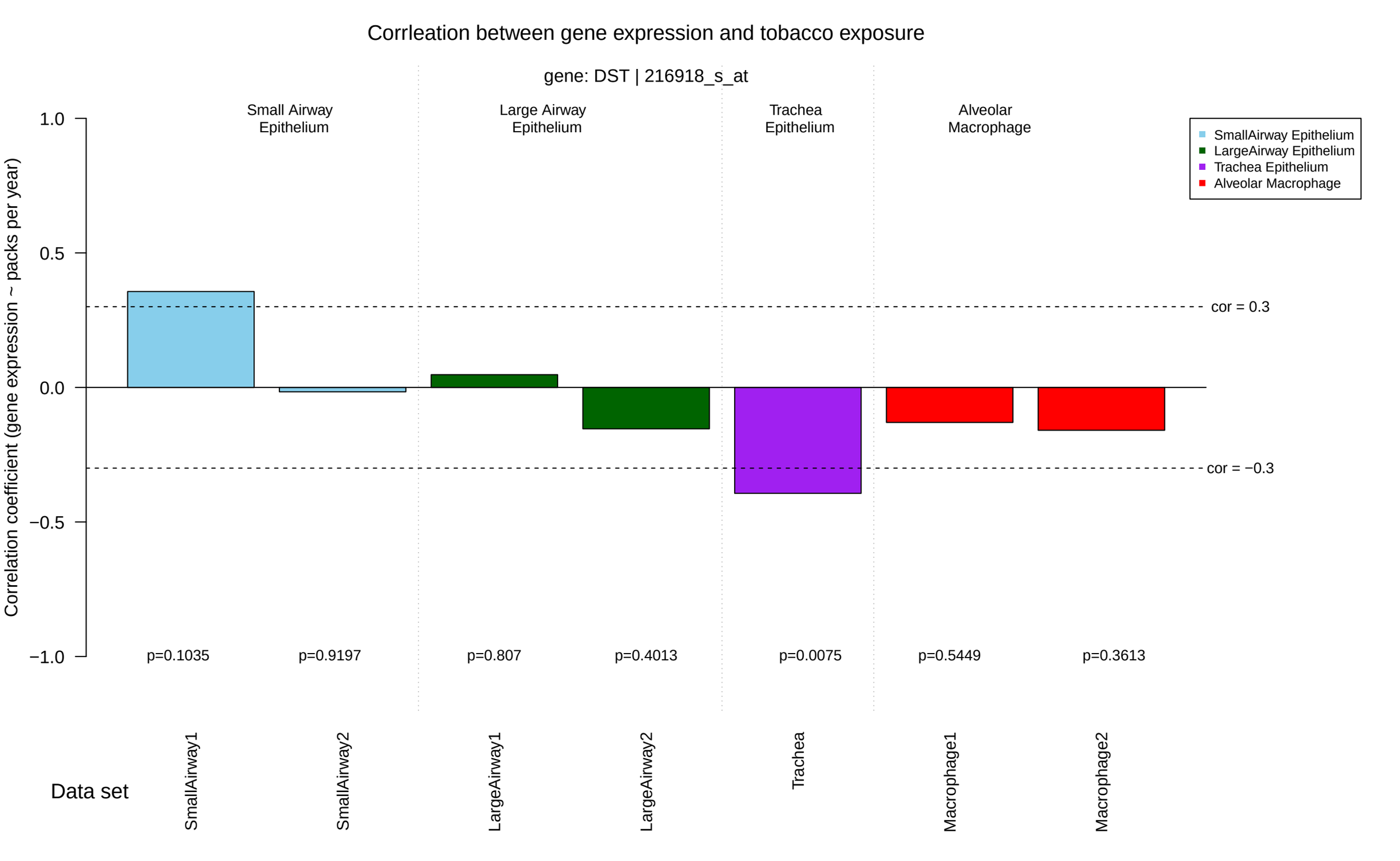

set: LargeAirway1 (p=0.8162)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.889)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0034)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1867)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6454)
DST | 216918_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DST | 216918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1035
p=0.9197
p=0.807
p=0.4013
p=0.0075
p=0.5449
p=0.3613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2